

Northwind Traders: Business Intelligence Report

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Introduction

Northwind Traders is a global gourmet food supplier that operates across multiple countries, selling products ranging from beverages to specialty foods. As part of an internal effort to improve visibility into business performance, this report presents a structured analysis of the company's sales data, covering the period from 2013 to 2015.

The goal was to build an executive dashboard that gives leadership a clear, at-a-glance view of three core areas: sales and revenue trends, product and category performance, and regional and people performance. The dataset used includes six interconnected tables — orders, order details, customers, products, employees, and shippers — which were modeled and analyzed using Power BI Desktop.

Data Overview

The dataset was sourced from the Northwind Traders Airtable database and contains the following tables:

****Orders**** — 830 total orders placed between July 2013 and May 2015

****Order Details**** — line-item breakdown of each order including quantity, unit price, and discount

****Customers**** — 91 unique customers across 21 countries

****Products**** — 77 products across 8 categories, of which 8 have been discontinued

****Employees**** — 9 sales employees based across the USA and UK

****Shippers**** — 3 shipping companies handling all deliveries

Before building the dashboard, the data was cleaned and structured in Power BI.

Relationships were established between all tables using a star schema, with the orders table at the center. Five DAX measures were created to power the visuals: Total Revenue, Total Orders, Average Order Value, Total Freight, and Discount Impact.

Dashboard Structure

The dashboard was built across three pages:

Page 1 — Sales & Revenue Overview

This page answers the question of how the business has performed over time. It includes KPI cards showing headline numbers, a line chart tracking monthly revenue trends, a bar chart

breaking down revenue by product category, and three slicers allowing the user to filter everything by year, country, and category.

Page 2 — Product & Category Intelligence

This page focuses on what is being sold. It includes a ranked bar chart of the top revenue-generating products, a pie chart showing the split between active and discontinued products, and a treemap giving a visual representation of category dominance.

Page 3 — Regional, Operational & People Performance

This page looks at where revenue is coming from and who is driving it. It includes a country-level revenue breakdown, an employee performance ranking, and a comparison of freight costs across the three shipping companies.

4. Key Findings

4.1 Revenue & Sales Trends

Total revenue across the full period stands at approximately \$1.27 million, generated from 830 orders. The average order value is \$1,530. Looking at monthly trends, revenue peaked in April 2015, with a noticeable dip during the summer months of 2014. There is a general upward trajectory heading into early 2015, suggesting the business was growing before the dataset ends.

4.2 Product & Category Performance

Beverages is the highest-grossing category by a significant margin, followed by Dairy Products and Confections. At the individual product level, Côte de Blaye stands out as the single highest revenue product, generating nearly \$150,000 on its own more than double the next best product.

Regarding discontinued products, 85.4% of total revenue comes from products that are still active. However, discontinued products still account for roughly 14.6% of historical revenue, which suggests that some of those products had a loyal customer base before being phased out. This is worth considering when making future discontinuation decisions.

4.3 Regional Performance

The United States is the largest market by revenue, followed by Germany and Austria. Together, these three countries account for a disproportionate share of total sales. Several European markets — including France, the UK, and Brazil, show moderate performance,

while smaller markets like Mexico and Switzerland contribute minimally. This concentration of revenue in a small number of countries presents both a risk and an opportunity. The business is heavily dependent on the US and German markets, meaning any disruption there would have an outsized impact on overall performance.

4.4 Employee Performance

Margaret Peacock is the top-performing sales employee by total revenue generated, followed closely by Janet Leverling and Nancy Davolio. The gap between the top three and the rest of the team is noticeable. Steven Buchanan and Michael Suyama sit at the bottom of the ranking, which could indicate smaller territories, fewer accounts, or an opportunity for additional support or training.

4.5 Shipping & Freight

United Package handles the highest freight volume in terms of cost, followed by Federal Shipping and Speedy Express. The difference between the three is not dramatic, suggesting a relatively even distribution of shipments. However, United Package consistently carries a higher cost load, which could be worth reviewing in future shipping contract negotiations.

Recommendation

Based on the analysis, the following actions are worth considering:

1. First, the business should look at replicating the conditions that drove the April 2015 revenue peak — whether that was a promotion, a seasonal factor, or a large client order — and see if it can be repeated or scaled.
2. Second, Beverages and Dairy Products should be prioritised in inventory and marketing planning given their dominant contribution to revenue.
3. Third, geographic expansion into mid-performing markets like France and Brazil could yield significant returns with relatively modest investment, given that those markets already have an established customer base.
4. Fourth, a structured performance review for lower-performing sales staff would help identify whether the gap is due to territory size, product mix, or skill gaps.

5. Finally, the freight cost distribution should be reviewed annually to ensure the business is getting competitive rates from its shipping partners.

Conclusion

This dashboard gives the Northwind Traders executive team a practical, interactive tool for monitoring business health across sales, products, and people. The insights surfaced through this analysis highlight both areas of strength and areas that deserve closer attention going into the next planning cycle. With filters for year, country, and category built in, the dashboard can be used on an ongoing basis — not just as a one-time report.

